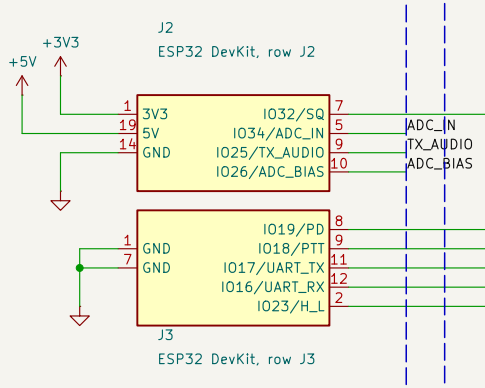
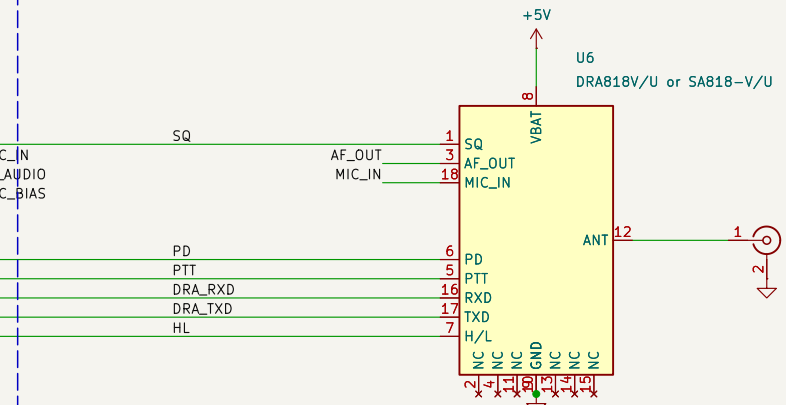
1 ESP32 DevKit



Its own pins are soldered into
the carrier (no sockets), about
3 mm gap. NodeMCU-32S / DevKitC V4
order, rows 22.86 mm apart:
J2 = left row (3V3 ... 5V)
J3 = right row (GND ... CLK)

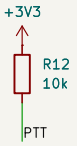
2 Radio module: DRA818V/U or SA818-V/U

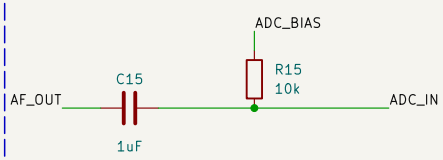


DRA_RXD = module RXD (IO17)
DRA_TXD = module TXD (IO16)
SQ is active low

H/L: floating = 1 W, low = 0.5 W.
IO23: output LOW or input,
never drive it HIGH.

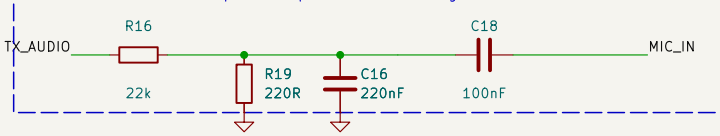
RX: AF_OUT -> 1uF -> IO34.
IO26 biases it to 1.75 V
through 10k (as the firmware expects).



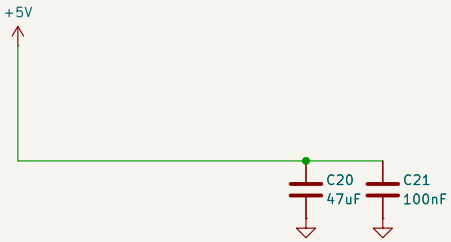


R12: PTT held high,
module stays in receive
while the ESP32 boots.

TX: IO25 (PDM) -> 22k / 220R divider with 220nF
low-pass (module wants about 10 mV) -> 100nF -> MIC_IN.
Audio values are first-pass; compare with the working board.



3 Radio supply



From the dev board's 5 V pin, straight to the module.
Module: 3.3-4.5 V, 0.75 A at TX; the dev board's
5 V pin is already diode-dropped (about 4.5 V).

Audio and power values are first-pass; verify on the bench
Same pin map as the firmware (GPIO16/17/18/19/25/26/32/34), plus GPIO23 for H/L

Sheet: /
File: rdio.kicad_sch

Title: rdio — ESP32 DevKit + DRA818/SA818 radio carrier

Size: A4	Date: 2026-11-05	Rev: A
KiCad E.D.A. 10.0.6		Id: 1/1